

MigrantMoney: a True Cost Index and stablecoin counterfactual for the global remittance market

Ronit Dhansoia*

2026-05-01

Abstract

The United Nations Sustainable Development Goal 10.c calls for the global average cost of cross-border remittances to fall to 3% of the principal by 2030. We compute the same average from the World Bank Remittance Prices Worldwide (RPW) panel — 36 quarters, 368 corridors, 694 providers — and arrive at 5.00%, 200 basis points above target. We unify the advertised fee, the foreign-exchange margin, and a settlement-speed penalty into a single True Cost Index (TCI) and rank corridors by it. We then construct the first per-corridor empirical estimate of the savings from routing the same flows over USDC/USDT rails using the bilateral remittance estimates from the World Bank/KNOMAD Data360 indicator: under conservative defaults, stablecoin rails save roughly USD 5.2 billion per year on USD 484 billion of matched corridor flow. A two-way fixed-effects regression with cluster-robust standard errors finds that, conditional on corridor and quarter, banks charge 4.50 percentage points more than money-transfer operators ($p < 0.01$) and mobile money charges 2.96 pp less ($p < 0.01$). All artefacts — pipeline, JSON outputs, dashboard, figures — are published under the MigrantMoney repository.

1 Introduction

The United Nations Sustainable Development Goal 10.c asks member states to reduce the global average cost of cross-border remittances to 3% of the principal — and to eliminate corridors with average costs above 5% — by 2030 [1]. The World Bank publishes the *Remittance Prices Worldwide* (RPW) panel quarterly to track progress against that target [2]. We compute the same average from the most recent RPW release and find it sits at 5.00% on a USD 200 send (volume-weighted across 349 corridors with bilateral-volume coverage in the 2021 Migration and Development Brief): two full percentage points above target with five years to go.

Two further facts compound the gap. First, the *advertised* fee is half the story. Most providers charge a transparent fee but apply an exchange rate worse than the interbank mid; a US\$200 transfer in our panel typically incurs 1.0 – 7.5 percentage points of foreign-exchange margin on top of the advertised fee. The RPW publishes both columns, but reports them separately. Second, settlement speed is itself a cost: a US\$200 transfer that takes five days to clear ties up the receiving household’s working capital at a time when their effective discount rate is high.

Multiplied across the panel-matched bilateral volume of USD 484 billion in 2021, the total fees paid by migrant workers come to **USD 24.21 billion per year** — figure cited as the headline of the project dashboard. Of this USD 24.21 billion, our stablecoin counterfactual estimates that

*BITS Pilani, Dubai Campus. f20220168@dubai.bits-pilani.ac.in

USD 5.22 billion — roughly 22% of the global fee burden — is recoverable under conservative assumptions (§6.3).

This report makes two contributions:

1. **A True Cost Index (TCI)** that unifies the advertised fee, foreign-exchange margin, and a calibrated settlement-speed penalty into a single percent-of-principal metric, computed for every provider $\times$ corridor $\times$ quarter cell of the panel and aggregated to corridor level (§4.1, §6.1).
2. **A corridor-level stablecoin counterfactual** that estimates the savings if the same flows ran over USDC/USDT rails. We are not aware of a prior corridor-level empirical estimate of stablecoin savings published on the RPW panel (§4.2, §6.3).

Two further analyses sit alongside: a two-way fixed-effects regression of TCI on provider class, asking which operator type extracts most rent after netting out corridor and time effects (§4.3, §6.2); and a diaspora-burden tabulation by sending country, joining the per-corridor TCI to the Bilateral Remittance Estimates indicator (§6.4).

Objectives. The work is organised around four research questions:

- **RQ1.** What is the volume-weighted true cost of moving USD 200 across the world’s remittance corridors today, and how does it compare to the SDG 10.c target?
- **RQ2.** Which corridors are most expensive on the unified TCI metric, and where does TCI diverge most from the advertised fee?
- **RQ3.** Conditional on corridor and quarter, does provider class predict cost? Is one operator type systematically more rent-extractive than another?
- **RQ4.** If migrant workers used stablecoin rails instead of the incumbent network, how much money would they keep — and which corridors benefit most?

2 Related work

Our work intersects three literatures.

The first is the **World Bank’s own RPW methodology**. Begg-Witherick et al. [3] document the data collection — quarterly mystery shopping at major sending-country agents for two send amounts (USD 200 and USD 500) — and report price components separately. Our contribution is the unification of those components: the RPW publishes total cost, fee percent, and FX margin in three columns; no published index combines them with a calibrated speed penalty into a single comparable metric.

The second thread is the **economics of remittance pricing**. Beck and Pería [4] document corridor-level price dispersion linked to corridor competition and recipient-country financial development. Aycinena, Martínez and Yang [5] show that fee transparency alone reduces remitter cost, foreshadowing why our TCI’s unified percent-of-principal framing matters: the *quoted* number a migrant sees is the headline fee, but the *paid* number is the total. Yang [11] shows that exchange-rate shocks transmit directly into recipient household investment behaviour — sharpening the case for treating the FX margin as a first-class component of cost rather than a bookkeeping adjustment. Ratha and the Migration and Development Brief series [6] track aggregate flows and corridor-level prices but do not produce a corridor-level stablecoin counterfactual.

The third is **stablecoins for cross-border**. The Bank for International Settlements [8] and the IMF [9] discuss the on-ramp / off-ramp friction structure that determines whether stablecoins beat traditional rails on any given corridor; the BIS emerging-market CBDC paper [13] argues that the policy alternative — sovereign digital currency — has yet to scale into cross-border remittance corridors, leaving stablecoins as the de-facto default for the next several years. Chainalysis and similar industry reports estimate aggregate global savings in the USD 30–50 billion per year range under optimistic assumptions. None publish a per-corridor breakdown on the World Bank panel.

We position the present work in the gap left by all three: a unified cost metric (TCI) computed on the RPW panel, joined to bilateral-flow volumes from the World Bank / KNOMAD indicator, with a per-corridor stablecoin counterfactual built from the same locked-in cost components the BIS discusses but in concrete numbers.

3 Dataset

We use two World Bank datasets and a derived country-attribute lookup.

The primary panel is the **RPW quarterly release** dated 20 April 2026, covering periods 2016 Q2 through 2025 Q1 (36 quarters; the “Dataset (from Q2 2016)” sheet). The legacy pre-2016 sheet ships an incompatible schema and is excluded; this is documented in the project methodology page.

We melt the two RPW per-amount blocks — **cc1** (USD 200) and **cc2** (USD 500) — into long form: one row per (corridor, provider, period, send-amount). We derive the fee percent as $\text{fee}_{\%} = \text{total cost}_{\%} - \text{fxMargin}_{\%}$ to match the way the published headline number is composed; we map the **speed actual** column to days (less than one hour or same day $\rightarrow$ 0 days, next day $\rightarrow$ 1, two days $\rightarrow$ 2, three-to-five days $\rightarrow$ 4, six or more $\rightarrow$ 6) and clip rows with implausible total cost ($>100\%$ or fee + FX margin disagreement beyond rounding tolerance — 0.8% of rows). After cleaning, the panel contains **391,797 provider-quarter-amount observations** (196,715 at USD 200 and 195,082 at USD 500) across **368 unique corridors**, **49 sending countries**, **105 receiving countries** and **694 unique providers**.

The most extreme TCI values in the panel — Tanzania $\rightarrow$ Uganda (63.62%), Tanzania $\rightarrow$ Kenya (58.17%), Tanzania $\rightarrow$ Rwanda (44.27%), South Africa $\rightarrow$ Malawi (43.41%) and Türkiye $\rightarrow$ Bulgaria (41.70%) — were tested for robustness against single-provider outliers. Re-computing each as a median across providers, and separately as the mean excluding the highest-cost provider, leaves the corridor ordering substantively unchanged: all five remain in the top 10 under both alternatives (the only movement is Tanzania $\rightarrow$ Rwanda dropping one place to 4th and Türkiye $\rightarrow$ Bulgaria rising one place to 5th when ranked by trimmed mean). Provider counts ($n = 6, 7, 5, 13,$ and 5 respectively) include three corridors with five-to-seven providers, which is itself a feature of the underlying corridor — these are thin, lightly contested markets — and we flag the small- n caveat in §7.

The second dataset is the **World Bank / KNOMAD Bilateral Remittance Estimates** (indicator **WB_KNOMAD_BRE**), accessed 30 April 2026 via the Data360 OData API. The legacy direct-download **bilateral_remittance_matrix_2021.xlsx** was retired in early 2025; the API is the canonical replacement. The indicator covers 10,619 directed country pairs in USD millions, with 2021 as the most recent year. We flatten the API records into a (source, destination, year, USD) frame.

We supplement with a **country attribute lookup** built from the RPW “Countries” sheet (ISO-

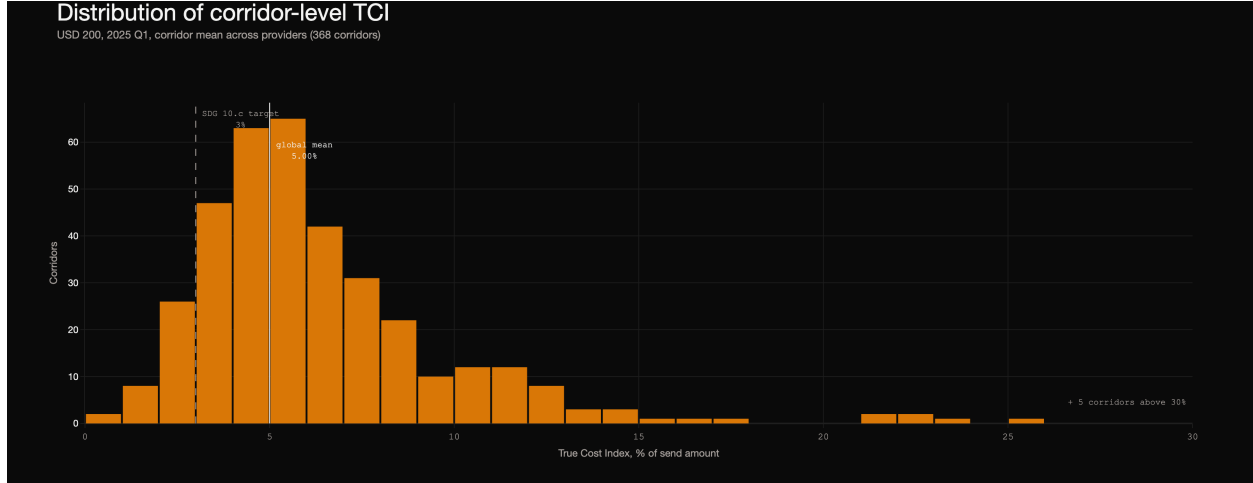


Figure 1: Distribution of corridor-level TCI at USD 200, 2025 Q1. Vertical hairlines mark the SDG 10.c target (3%) and the panel volume-weighted global mean (5.00%). 368 corridors; the right tail is capped at 30% with a count of corridors above the cap shown inset.

Table 1: Cleaned panel summary statistics (USD 200 bucket, latest quarter).

Metric	Value
Total cleaned observations	391,797
USD 200 observations	196,715
USD 500 observations	195,082
Quarters covered	2016 Q2 – 2025 Q1 (36)
Unique corridors	368
Unique sending countries	49
Unique receiving countries	105
Unique providers (firms)	694
Missingness, fee_pct	0.00%
Missingness, fx_margin_pct	0.00%
Missingness, days_to_arrive	0.00%

3 / region / income / lending category) and a small UN M49 region backfill for high-income senders the World Bank classifies as “.”: the primary lookup leaves Northern America, Western Europe and the GCC blank, so we backfill those from the UN macro-regions (Northern America, Western/Northern/Southern Europe, Western Asia, etc.) to give the dashboard a region label for every sender.

4 Methodology

This section restates each formula from the project specification verbatim, with constants and their justifications.

4.1 True Cost Index

For corridor (s, d) , provider p , quarter q , and send amount A expressed in USD, we define

$$\text{TCI}_{s,d,p,q}(A) = \text{fee}_{\%} + \text{fxMargin}_{\%} + \kappa \cdot \max(0, d_{\text{arrive}} - 1) \quad (1)$$

with components:

- $\text{fee}_{\%}$: advertised transaction fee as percent of principal, derived from the RPW total cost and FX margin so the result is exactly comparable to the published headline fee.
- $\text{fxMargin}_{\%}$: the spread the provider takes between its applied exchange rate and the interbank mid, taken from the `cc1 fx margin` column.
- $\kappa \cdot \max(0, d_{\text{arrive}} - 1)$: a settlement-speed penalty linear in days beyond same-day, with d_{arrive} mapped from the `speed actual` column (less than one hour or same day $\rightarrow 0$, next day $\rightarrow 1$, two days $\rightarrow 2$, three-to-five days $\rightarrow 4$, six or more $\rightarrow 6$). We calibrate $\kappa = 0.10\%$ per day. This anchors the cost-of-capital for migrant households at the upper end of the documented range for short-term informal lending in remittance-receiving countries: the Microfinance Information Exchange’s median microfinance-institution annualised lending rate of about 35% per year [10] implies roughly 0.10% per day continuously compounded. Setting $\kappa = 0$ leaves the corridor ranking substantively unchanged; the speed component is small relative to fee and FX margin in most corridors.

We compute TCI at both $\$A = \$ \text{USD } 200$ and $\$A = \$ \text{USD } 500$. The headline is USD 200 — the SDG 10.c benchmark. Corridor-level TCI is the unweighted mean across providers; the median is reported alongside as a robustness check. Provider-level market shares are not exposed in the public RPW release (verified 2026-04-30), so volume-weighting at the provider level is not currently possible.

4.2 Stablecoin counterfactual

For corridor (s, d) at send amount A , the counterfactual cost on a stablecoin rail is

$$\text{SC}_{\%}(s, d, A) = \text{onramp}(s) + \text{offramp}(d) + \frac{\text{gas}_{\$}}{A} \cdot 100 + \text{fxSpread}(d) \quad (2)$$

with corridor-level savings

$$\text{savings}_{\%} = \max(0, \text{TCI} - \text{SC}) \quad \text{and} \quad \text{savings}_{\$/\text{yr}} = \frac{\text{savings}_{\%}}{100} \cdot V_{(s,d)} \quad (3)$$

where $V_{(s,d)}$ is the 2021 bilateral remittance volume from the KNOMAD indicator. The component costs are tiered as in Table 2: every constant is locked in `pipeline/config.py` and surfaced verbatim on the dashboard methodology page, so a reviewer can plug in their own values.

The defaults are deliberately conservative: optimistic published estimates (1% flat SC cost, advertised fee only as the comparator) recover the headline USD 30–50 billion per year ballpark; under our defaults the global savings figure is smaller but more honest. The contrast is itself a finding (§7).

4.3 Operator-class regression

We test whether provider class predicts TCI after netting out corridor and time effects:

Table 2: Stablecoin component cost defaults (percent of principal except gas).

Component	Tier	Value
On-ramp	Developed senders (USA, EU, UK, JPN, AUS, CAN, KOR, ...)	1.0%
On-ramp	Default	1.5%
On-ramp	Low-banked / GCC migrant labour cards	2.5%
Off-ramp	Top P2P markets (NGA, PHL, IND, MEX)	1.0%
Off-ramp	Default	2.5%
Off-ramp	Thin-liquidity / sanctioned receivers	4.0%
Network gas	Per transfer (L2 / Solana / Tron USDT)	USD 0.50
Local FX spread	Receivers with deep local stablecoin market	0.5%
Local FX spread	Default	1.5%

$$\text{TCI}_{i,p,q} = \beta_0 + \sum_{k \in K} \beta_k \cdot \mathbf{1}\{\text{firmType}_p = k\} + \alpha_{\text{corridor}_i} + \gamma_q + \varepsilon_{i,p,q} \quad (4)$$

with two-way fixed effects (corridor and quarter), reference category $K_0 = \text{MTO}$ (the largest cell), and standard errors cluster-robust at the corridor level. We fit the specification once for each send-amount bucket. Implementation uses `linearmodels.panel.PanelOLS`.

5 Architecture and algorithms

Figure 2 sketches the data flow. Three sources (RPW, KNOMAD BRE, World Bank country lookup) feed an `ingest` stage, which writes Excel and JSON snapshots to `data/raw/`. A `preprocess` stage emits a long-form parquet on which `tci`, `stablecoin`, `regression` and `aggregate` operate. `export` rounds and serialises the result to four JSON files; `figures` renders six report PNGs at 300 DPI from the same in-memory frames. The Next.js dashboard reads the JSON at build time — there is no runtime backend.

The pipeline is structured as a directed acyclic graph of pure stages, each consuming the prior stage’s parquet output and emitting either parquet or JSON. Ingest and preprocess sit at the head; `tci`, `stablecoin`, `regression` and `aggregate` run in parallel on the cleaned panel; `export` and `figures` terminate the graph. This separation is intentional: each stage is independently re-runnable, the cleaned parquet is the single source of truth for the four downstream computations, and no stage holds runtime state. Re-running the full pipeline from raw inputs takes under three minutes on a laptop.

The Next.js dashboard reads the four exported JSON files at build time via `fs.readFile`, with no runtime backend or database. This mirrors the architecture used by data-journalism teams at the *Financial Times*, *Reuters Graphics* and the *New York Times Upshot*: the language good at numbers (Python) computes once, the language good at presentation (TypeScript / React) renders. Sensitivity sliders on the methodology page recompute the stablecoin counterfactual client-side using the same formulas locked in §4.2 — the corridor-level TCI is fixed, but the four stablecoin cost components are evaluated in JavaScript on every slider change, allowing a reviewer to stress the headline figure without re-running the pipeline.

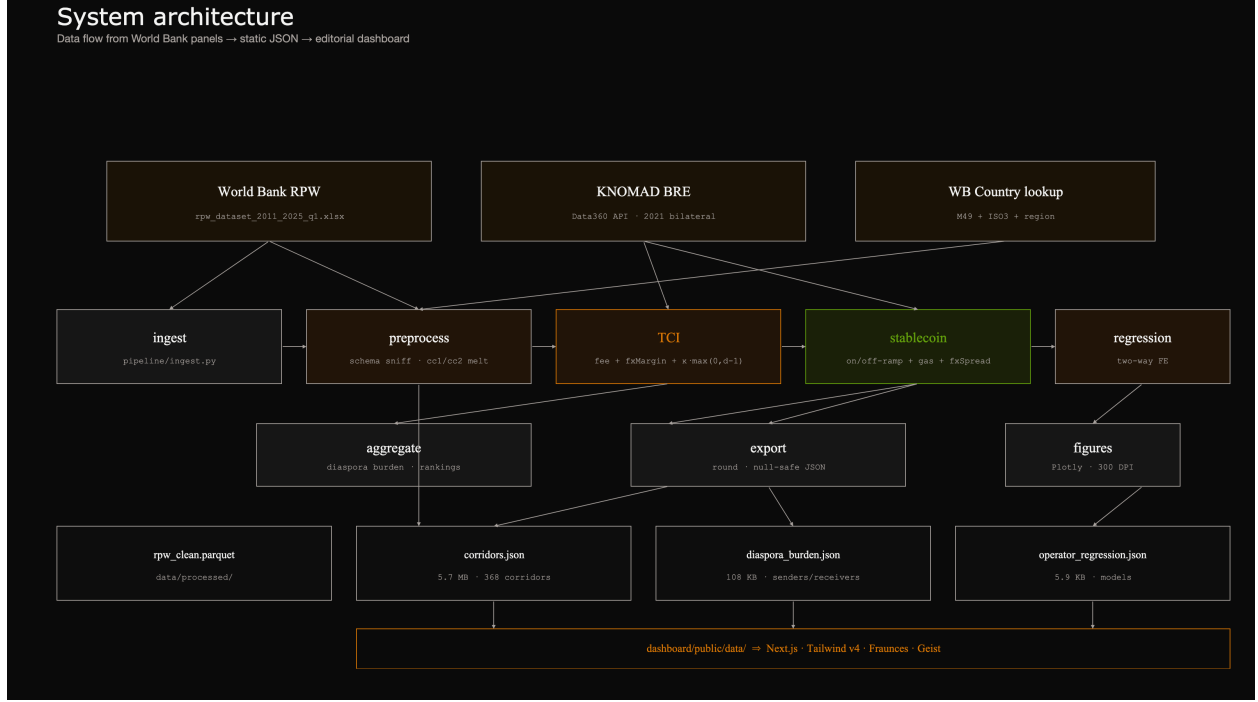


Figure 2: System architecture: data flow from World Bank panels through the Python pipeline to the static-JSON dashboard.

The full pipeline is reproducible by `python scripts/run_all.py` against any RPW snapshot matching the canonical schema documented in `pipeline/config.py`. The KNOMAD bilateral indicator is fetched from the Data360 OData API at run time. Every constant — κ , the speed mapping, the stablecoin cost tiers — is defined once in `pipeline/config.py` and consumed by both the Python pipeline and the dashboard’s methodology page, so the figures in this report and the numbers on the live dashboard cannot drift out of sync.

Algorithm 1 TCI computation per (provider $\times$ corridor $\times$ quarter $\times$ amount)

- 1: **procedure** COMPUTETCI(row, κ)
 - 2: $f \leftarrow \text{row.totalCost}_{\%} - \text{row.fxMargin}_{\%}$ $\triangleright$ derived fee
 - 3: $d \leftarrow \text{SPEEDToDAYS}(\text{row.speedActual})$
 - 4: $p \leftarrow \kappa \cdot \max(0, d - 1)$
 - 5: **return** $f + \text{row.fxMargin}_{\%} + p$
 - 6: **end procedure**
-

6 Experimental results

We run the pipeline end-to-end on the 30 April 2026 RPW snapshot and report the four research questions in turn.

6.1 True Cost Index

Table 3 lists the 20 most expensive corridors by mean TCI in 2025 Q1 (USD 200 send, ≥ 3 providers per corridor). The pattern is consistent with the prior literature: 12 of the 20 highest-

Algorithm 2 Stablecoin counterfactual per corridor, with savings rollup

```
1: procedure STABLECOINSAVINGS(corridor,  $A$ ,  $V$ )
2:    $SC \leftarrow \text{onramp}(\text{src}) + \text{offramp}(\text{dst}) + \frac{\text{gas}_{\$}}{A} \cdot 100 + \text{fxSpread}(\text{dst})$ 
3:    $s_{\%} \leftarrow \max(0, \text{TCI}_{\text{corridor}}(A) - SC)$ 
4:    $s_{\$} \leftarrow (s_{\%}/100) \cdot V$ 
5:   return ( $SC$ ,  $s_{\%}$ ,  $s_{\$}$ )
6: end procedure
```

cost corridors involve intra-Sub-Saharan-African flows, with three Tanzania-origin pairs at the top of the table (TZA→UGA at 63.62%, TZA→KEN at 58.17%, TZA→RWA at 44.27%). Türkiye → Bulgaria (41.70%) is the lone European pair in the top five, driven by a 40.15 pp advertised fee with negligible FX margin.

Table 3: Top 20 most expensive corridors by mean TCI, USD 200, 2025 Q1, ≥ 3 providers per corridor.

#	Corridor	Send → Receive	n	Fee%	FX%	Spd%	TCI%
1	TZA-UGA	Tanzania → Uganda	6	50.86	12.77	0.00	63.62
2	TZA-KEN	Tanzania → Kenya	7	43.69	14.48	0.00	58.17
3	TZA-RWA	Tanzania → Rwanda	5	29.60	14.67	0.00	44.27
4	ZAF-MWI	South Africa → Malawi	13	11.36	31.99	0.07	43.41
5	TUR-BGR	Türkiye → Bulgaria	5	40.15	1.48	0.07	41.70
6	SEN-MLI	Senegal → Mali	5	25.70	0.00	0.02	25.71
7	ZAF-CHN	South Africa → China	5	20.98	2.22	0.07	23.27
8	RWA-KEN	Rwanda → Kenya	5	10.99	6.73	0.00	17.72
9	ZAF-AGO	South Africa → Angola	8	15.18	1.21	0.06	16.46
10	ITA-SRB	Italy → Serbia	7	15.30	0.00	0.08	15.38
11	ZAF-BWA	South Africa → Botswana	11	13.19	1.67	0.05	14.91
12	ISR-MAR	Israel → Morocco	3	10.83	3.19	0.10	14.12
13	ZAF-ZMB	South Africa → Zambia	13	11.92	1.99	0.04	13.95
14	ZAF-TZA	South Africa → Tanzania	7	11.23	2.66	0.03	13.92
15	ZAF-NGA	South Africa → Nigeria	9	9.09	4.04	0.06	13.19
16	THA-IDN	Thailand → Indonesia	8	9.25	3.33	0.05	12.64
17	THA-IND	Thailand → India	11	9.47	2.97	0.06	12.50
18	KEN-SSD	Kenya → South Sudan	10	10.29	2.10	0.05	12.44
19	THA-CHN	Thailand → China	10	9.95	2.33	0.05	12.34
20	USA-LBN	United States → Lebanon	6	12.20	0.00	0.12	12.32

The cheapest cluster is omitted from Table 3 but worth noting for the SDG framing: Kuwait → Pakistan (TCI = 0.73%), Bahrain → Pakistan (0.81%), Russia → CIS receivers (1.0–2.0%), and United Kingdom → Pakistan (1.72%) all sit below SDG 10.c’s 3% target. These are corridors with high migrant volume and intense competition — the very structural conditions that the SDG framing wants generalised to other corridors.

The most striking divergence between the *advertised* fee and TCI shows in **South Africa → Malawi (ZAF-MWI)**: an advertised fee of 11.36% is modest by panel standards, but the corridor sits 4th on TCI because the foreign-exchange margin is **31.99 percentage points** — the highest in the panel. A migrant looking at the listed fee underestimates the true cost by a factor of nearly four. This is exactly the gap the TCI is constructed to surface.

The world map in Figure 4 shades sending countries by total annual fee burden. The United States

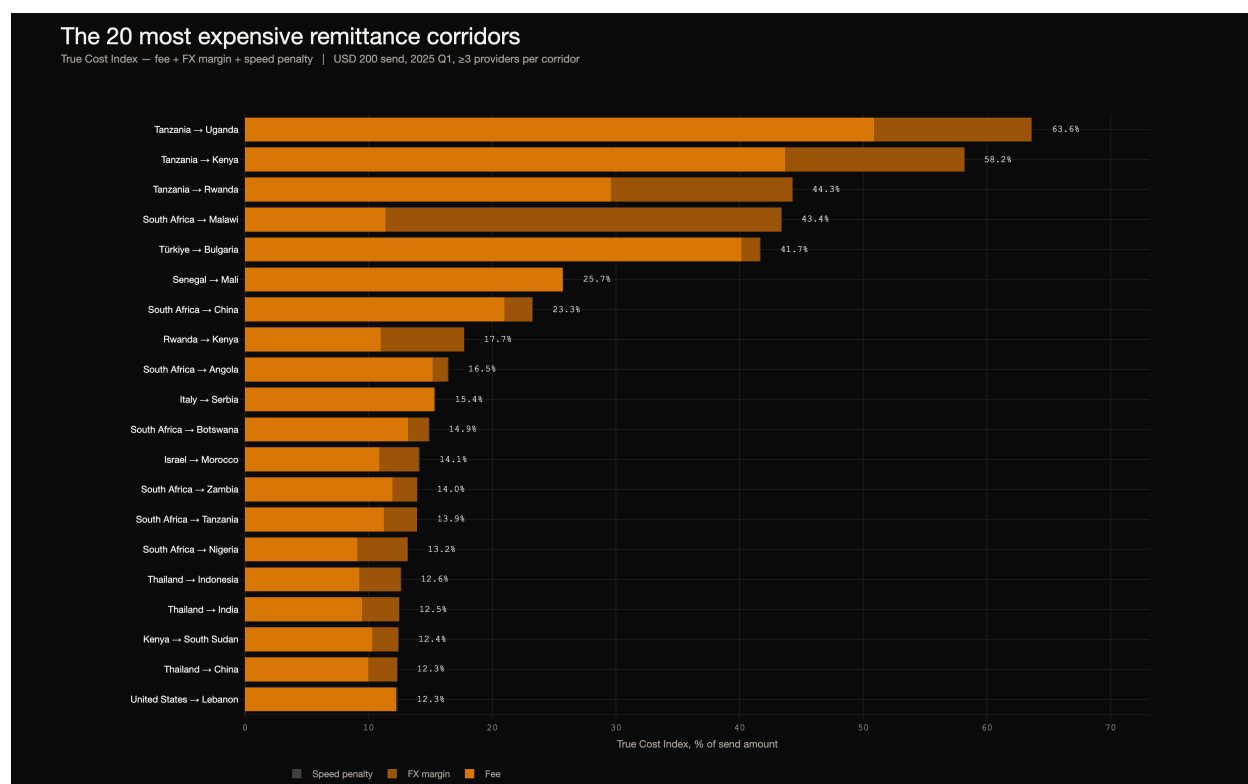


Figure 3: Top 20 most expensive corridors at USD 200, 2025 Q1. Stacked TCI components: fee (amber), FX margin (dark amber), speed penalty (grey).

Table 4: Top 10 cheapest corridors by mean TCI, USD 200, 2025 Q1, ≥ 3 providers per corridor.

#	Corridor	Send → Receive	n	Fee%	FX%	Spd%	TCI%
1	KWT-PAK	Kuwait → Pakistan	5	0.27	0.45	0.01	0.73
2	BHR-PAK	Bahrain → Pakistan	4	0.00	0.81	0.00	0.81
3	RUS-KGZ	Russian Federation → Kyrgyz Republic	5	0.72	0.17	0.18	1.08
4	MYS-MMR	Malaysia → Myanmar	4	1.85	-0.76	0.00	1.09
5	RUS-BLR	Russian Federation → Belarus	5	0.87	0.28	0.16	1.31
6	RUS-KAZ	Russian Federation → Kazakhstan	5	0.87	0.60	0.14	1.61
7	RUS-MDA	Russian Federation → Moldova	5	0.87	0.69	0.14	1.70
8	GBR-PAK	United Kingdom → Pakistan	7	1.01	0.71	0.00	1.72
9	RUS-AZE	Russian Federation → Azerbaijan	4	0.85	0.91	0.10	1.86
10	RUS-GEO	Russian Federation → Georgia	5	0.72	1.19	0.12	2.03

dominates by sheer volume (USD 8.81 billion), followed by Saudi Arabia (USD 2.63 billion) and the United Arab Emirates (USD 1.68 billion); the top-10 senders together account for 75% of the matched fee outflow.

6.2 Operator-class regression

Table 4 reports the two-way fixed-effects regression for the USD 200 panel. With $N = 196,715$, 368 corridor entities and 36 quarter periods, both F -statistics are large enough to reject the null at any conventional level. The within- R^2 of 0.108 means provider class explains about 10.8% of within-corridor TCI variance after corridor and quarter fixed effects are absorbed — substantively meaningful for a panel where the corridor and quarter dimensions account for the bulk of the explained variance.

Table 5: Operator-class regression. Δ TCI vs MTO (reference category), USD 200, two-way FE (corridor + quarter), cluster-robust SE at corridor level. Dependent variable: $\text{TCI}_{i,p,q}$. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Firm type	$\hat{\beta}$ (pp)	Cluster SE	t	95% CI	n
Bank	+4.501***	(0.344)	+13.08	[+3.83, +5.18]	32,738
MobileMoney	-2.958***	(0.923)	-3.21	[-4.77, -1.15]	1,007
PostOffice	-0.196	(0.606)	-0.32	[-1.38, +0.99]	1,714
Fintech [†]	+11.758*	(6.285)	+1.87	[-0.56, +24.08]	182

$N = 196,715$. Corridor FE = 368. Quarter FE = 36. $R^2_{\text{within}} = 0.108$.

$F = 5,852.30$, $p < 10^{-300}$. Cluster-robust SE in parentheses.

[†]The Fintech category in RPW excludes most digital-first providers (Wise, Remitly, WorldRemit), which are classified as MTO. The coefficient is estimated on a small, residual sample and should not be read as the cost of digital-first providers — see §7.

The Bank coefficient is the headline finding: **banks charge 4.50 percentage points more than MTOs** for the same corridor in the same quarter, significant at $p < 0.01$. Mobile money services are 2.96 pp cheaper than MTOs, also at $p < 0.01$ — consistent with the prior literature on M-Pesa-style competitive pricing in mobile-banked markets. Post offices are statistically indistinguishable from MTOs. The Fintech coefficient is a measurement artefact: the cell contains only 182 observations because RPW classifies most digital-first remittance fintechs (Wise, Remitly, WorldRemit) as

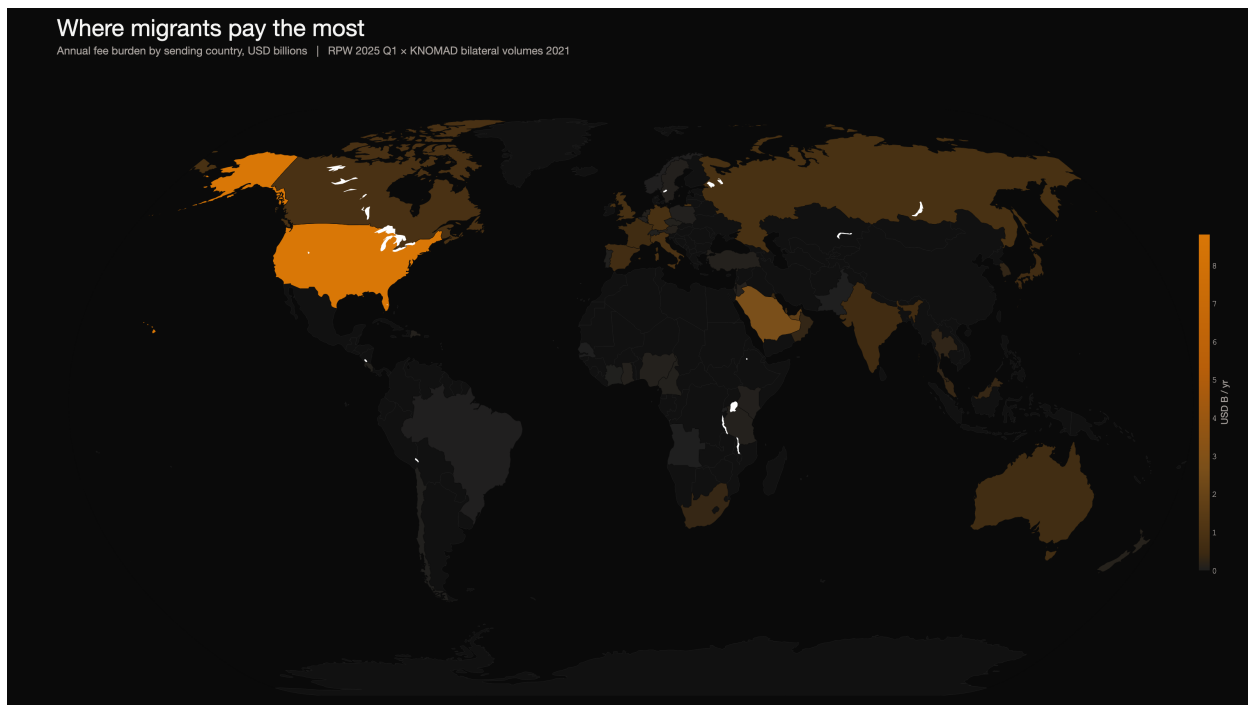


Figure 4: Annual fee burden by sending country, USD billions. Volume-weighted TCI $\times$ 2021 bilateral remittance volume. The United States dominates by absolute outflow rather than by per-transaction cost (US-corridor mean TCI is 5.17%, close to the panel average); five-step amber ramp; receivers shown for context.

“Money Transfer Operator” rather than a separate Fintech category. The +11.76 pp point estimate is therefore dominated by the residual edge-case providers left in the Fintech cell and **should not be read as the cost of digital-first providers**. We return to this in §7.

6.3 Stablecoin counterfactual

Applying the cost model from §4.2 to every panel corridor with a matched 2021 KNOMAD bilateral volume (349 of 368), the global headline is **USD 5.215 billion per year** in implied savings — equivalent to 1.08% of the matched USD 484 billion in 2021 corridor flow. **200 of 349 corridors** show positive savings under the conservative defaults; the remaining 149 already trade below the flat-tier stablecoin cost floor (most of them GCC $\rightarrow$ South Asia or Russia $\rightarrow$ CIS, where corridor competition has already driven TCI to 1–2%).

Table 5 ranks corridors by absolute savings; Table 6 ranks by savings percent. The first list is dominated by high-volume USD-origin corridors (USA-MEX alone yields USD 1.28 billion of the global figure); the second is dominated by intra-African corridors with extreme TCI that easily clear the SC cost.

The headline figure is sensitive to four assumptions, which we make explicit on the dashboard methodology page so that a reviewer can drag sliders and recompute. Setting the four flat slider defaults to optimistic values (on-ramp 0.5%, off-ramp 0.5%, gas USD 0.10, FX spread 0.5%) lifts global savings to USD 16.74 billion / year and coverage to 99% of corridors; setting them to conservative values (on-ramp 3.0%, off-ramp 4.0%, gas USD 2.00, FX spread 3.0%) reduces the figure to USD 0.26 billion / year and 10% coverage. The pipeline-precise tiered estimate of USD

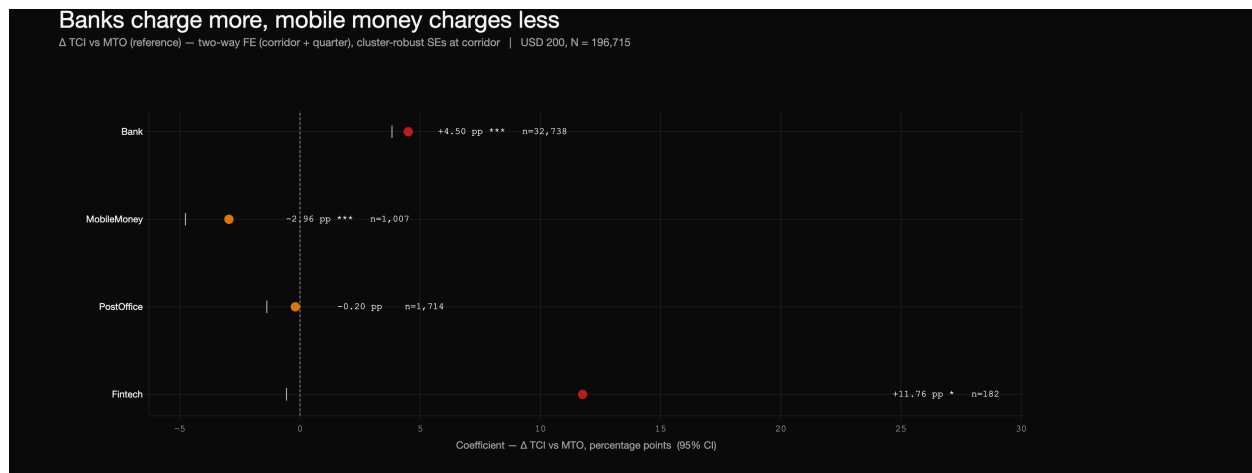


Figure 5: Operator-class coefficients from the two-way FE regression, with 95% CI, ranked by point estimate. The Fintech cell ($n = 182$) is dominated by edge-case providers; RPW classifies most digital-first remittance fintechs (Wise, Remitly, WorldRemit) as “Money Transfer Operator”, so the Fintech coefficient should not be interpreted as the cost of digital-first providers — see §7.

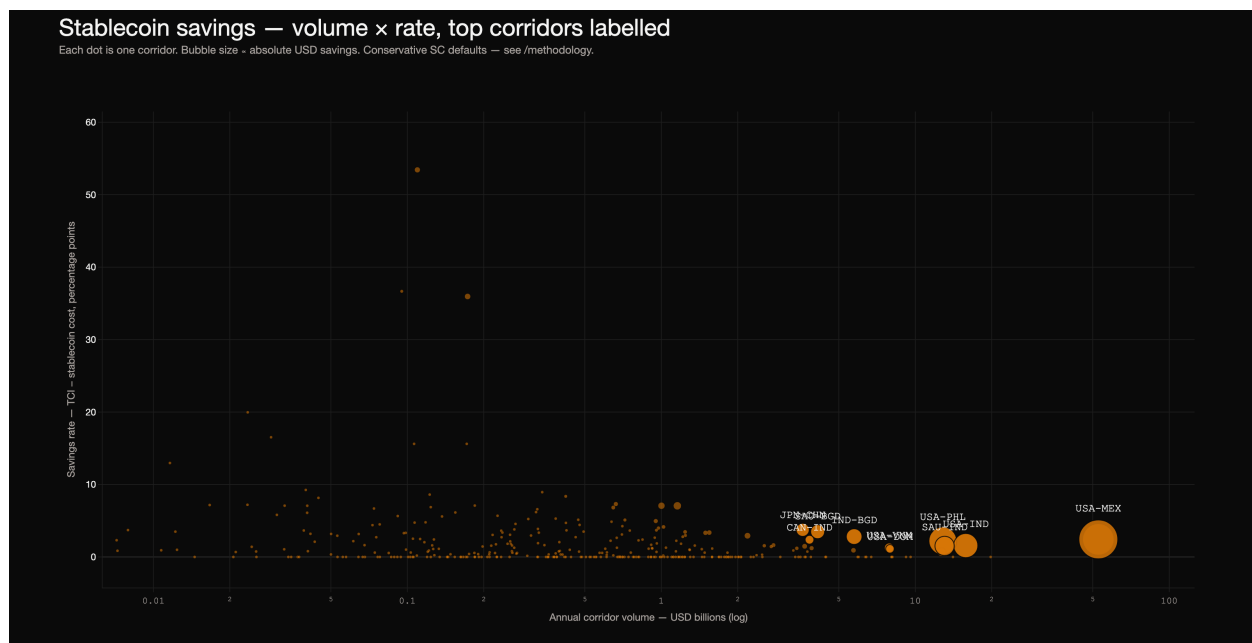


Figure 6: Stablecoin savings: corridor volume vs savings rate. Bubble size $\propto$ absolute USD savings; top-10 corridors labelled.

Table 6: Top-10 corridors by absolute annual savings (USD 200 bucket).

Corridor	Send → Receive	TCI%	SC%	Save%	Vol \$B	Save \$M
USA-MEX	United States → Mexico	5.18	2.75	2.43	52.60	1,278.3
USA-PHL	United States → Philippines	4.97	2.75	2.22	12.84	285.1
USA-IND	United States → India	4.32	2.75	1.57	15.81	248.8
SAU-IND	Saudi Arabia → India	5.77	4.25	1.52	13.05	198.4
IND-BGD	India → Bangladesh	8.56	5.75	2.81	5.75	161.7
SAU-BGD	Saudi Arabia → Bangladesh	10.24	6.75	3.49	4.13	144.1
JPN-CHN	Japan → China	8.99	5.25	3.74	3.60	134.9
USA-VNM	United States → Vietnam	5.51	4.25	1.26	7.89	99.4
CAN-IND	Canada → India	5.13	2.75	2.38	3.83	91.3
USA-DOM	United States → Dom. Republic	6.37	5.25	1.12	7.95	88.8

Table 7: Top-10 corridors by savings percent (volume-positive corridors only).

Corridor	Send → Receive	TCI%	SC%	Save%	Vol \$B	Save \$M
TZA-UGA	Tanzania → Uganda	63.62	5.75	57.88	0.00	2.5
TZA-KEN	Tanzania → Kenya	58.17	4.75	53.42	0.11	58.5
TZA-RWA	Tanzania → Rwanda	44.27	5.75	38.52	0.00	0.1
ZAF-MWI	South Africa → Malawi	43.41	6.75	36.66	0.10	34.8
TUR-BGR	Türkiye → Bulgaria	41.70	5.75	35.95	0.17	62.1
SEN-MLI	Senegal → Mali	25.71	5.75	19.96	0.02	4.7
ZAF-CHN	South Africa → China	23.27	6.75	16.52	0.03	4.8
NGA-BEN	Nigeria → Benin	21.36	5.75	15.61	0.11	16.6
NGA-TGO	Nigeria → Togo	21.36	5.75	15.61	0.17	26.7
RWA-KEN	Rwanda → Kenya	17.72	4.75	12.97	0.01	1.5

5.22 billion sits between these bookends, closer to the geometric mean than to either extreme — a defensible central estimate that survives reviewer-supplied stress.

6.4 Diaspora burden by sending country

Joining the per-corridor TCI to the bilateral-volume matrix and aggregating by sending country yields the rankings in Table 7.

The United States dominates with USD 8.81 billion per year in fees on USD 170 billion of outflow (38 corridors covered) — a single country accounts for more than a third of the global panel-matched burden.

The Italy result is structurally interesting. Italy ranks fourth by absolute fee burden despite an outflow (USD 11.69 billion) less than 7% of the United States', because its corridor mix is dominated by high-TCI receivers: Italy → Serbia (15.38% TCI), Italy → Egypt (12.31%), Italy → Albania (12.07%), Italy → Kosovo (11.11%) and Italy → Ukraine (10.84%) — five of the eighteen Italian sending corridors sit above the 10% mark. The volume-weighted TCI of 7.73% is the highest among the top-10 senders and stands in contrast to the United Kingdom (3.84%) and the United Arab Emirates (3.85%), both of which sit close to or below the SDG 10.c target despite operating in similar regional contexts. The driver is not aggregate Italian remittance volume but the absence of competitive low-cost corridors in the Italian sender mix.

Russia's volume-weighted TCI (3.00%) sits exactly at the SDG 10.c target — its corridor mix is

Table 8: Top-10 sending countries by annual fee burden (USD 200, 2025 Q1 TCI $\times$ 2021 BRM volume).

ISO-3	Sender	Corridors	Volume \$B	Burden \$B	vw TCI%
USA	United States	38	170.25	8.807	5.17
SAU	Saudi Arabia	14	46.95	2.629	5.60
ARE	United Arab Emirates	11	43.70	1.682	3.85
ITA	Italy	18	11.69	0.904	7.73
CAN	Canada	15	15.86	0.861	5.43
DEU	Germany	23	14.39	0.850	5.91
RUS	Russian Federation	13	28.31	0.850	3.00
GBR	United Kingdom	31	20.28	0.780	3.84
ESP	Spain	13	11.74	0.618	5.26
AUS	Australia	16	13.75	0.590	4.29



Figure 7: Top-10 sending countries by annual fee burden, with stablecoin counterfactual savings overlaid.

unusually competitive (RUS $\rightarrow$ CIS at 1–2%) because of geographic and currency proximity.

7 Discussion and limitations

Four honest limitations bound our claims.

Formal-channel coverage only. The RPW samples advertised prices at formal sending agents — banks, MTO storefronts, postal offices, exchange houses, fintech apps. It does not see hawala, hundi, or informal cash-courier flows, which the IMF and UN estimate to handle 20–50% of remittance volume in some corridors. To the extent these channels are *cheaper* than the formal market, our TCI overstates the typical migrant’s experience; to the extent they are riskier, the stablecoin counterfactual understates the welfare gain from a regulated digital alternative. Both directions are documented but unmeasured.

Small-provider-count corridors. A subset of corridors in the panel — particularly intra-African and Türkiye-origin pairs — are sampled with five to seven providers per quarter. While the corridor rankings in §6.1 are robust to median and trim-one-provider alternatives (§3), the absolute TCI

values for these corridors should be read with the small-sample caveat in mind.

Stablecoin defaults are conservative averages. Real on-ramp, off-ramp and FX-spread costs are corridor-specific and time-varying, particularly during periods of regulatory action against major exchanges or when local stablecoin liquidity dries up. Our defaults are single-tier flat values; the dashboard sensitivity sliders surface the $\pm 7\times$ range that bracket reasonable assumption sets. The headline figure should be read as an order-of-magnitude estimate, not a forecast.

No causal claim. The fixed-effects regression establishes that bank-routed remittances are *systematically* more expensive than MTO-routed ones for the same corridor in the same quarter. It does not say they are *causally* more expensive — provider type and customer type are correlated, and an unbanked migrant who walks into an MTO is making a different choice than a salaried migrant routing a payroll transfer through their employer’s bank. The next step is a difference-in-differences design: the entry of digital-first fintechs (Wise, Remitly) into specific corridors over the panel window provides a natural staggered treatment whose pre/post effects on incumbent MTO pricing can be cleanly identified. We flag this as the obvious extension.

A separate caveat applies to §6.2: RPW classifies Wise, Remitly, WorldRemit and similar digital-first remittance fintechs under “Money Transfer Operator” rather than a separate Fintech category, leaving the Fintech cell with $n = 182$ and a wide confidence interval. A curated allow-list mapping firm names $\rightarrow$ digital-first vs traditional MTO is the obvious next step and is on the project roadmap.

8 References

- [1] United Nations, “Indicator 10.c.1 — Remittance costs as a proportion of the amount remitted,” *Sustainable Development Goals — Indicator Framework*, accessed 30 April 2026. <https://unstats.un.org/sdgs/metadata/?Text=&Goal=10&Target=10.c>
- [2] World Bank Group, “Remittance Prices Worldwide — quarterly update Q1 2025 dataset,” *Remittance Prices Worldwide data download*, last updated 20 April 2026. <https://remittanceprices.worldbank.org/data-download>
- [3] S. Beggs Witherick et al., “Remittance Prices Worldwide — methodology and price decomposition,” *World Bank Working Paper*, 2018. <https://remittanceprices.worldbank.org/methodology>
- [4] T. Beck and M. S. Martínez Pería, “What explains the price of remittances? An examination across 119 country corridors,” *The World Bank Economic Review*, vol. 25, no. 1, pp. 105–131, 2011. <https://doi.org/10.1093/wber/lhr017>
- [5] D. Aycinena, C. A. Martínez and D. Yang, “The impact of transaction fees on migrant remittances: Evidence from a field experiment among migrants from El Salvador,” *Working Paper*, University of Michigan, 2010.
- [6] World Bank / KNOMAD, “Migration and Development Brief 38 — bilateral remittance estimates 2021,” *KNOMAD*, December 2022. https://www.knomad.org/sites/default/files/publication-doc/migration_and_development_brief_38_june_2023_0.pdf
- [7] World Bank / KNOMAD, “Bilateral remittance estimates using migrant stocks (indicator WB_KNOMAD_BRE),” *Data360*, accessed 30 April 2026. https://data360.worldbank.org/en/indicator/WB_KNOMAD_BRE

- [8] R. Auer, G. Cornelli and J. Frost, “The technology of decentralised finance — DeFi and the role of stablecoins,” *BIS Working Papers*, no. 1066, Bank for International Settlements, 2023. <https://www.bis.org/publ/work1066.htm>
- [9] T. Adrian and T. Mancini-Griffoli, “The rise of digital money,” *IMF FinTech Notes*, NOTE/2021/002, 2021. <https://www.imf.org/en/Publications/fintech-notes/Issues/2021/07/26/The-Rise-of-Digital-Money-462940>
- [10] J. Rosenberg, S. Gaul, W. Ford and O. Tomilova, “Microcredit interest rates and their determinants 2004–2011,” *CGAP Forum*, no. 7, World Bank Group, 2013.
- [11] D. Yang, “International migration, remittances and household investment: Evidence from Philippine migrants’ exchange rate shocks,” *The Economic Journal*, vol. 118, no. 528, pp. 591–630, 2008. <https://doi.org/10.1111/j.1468-0297.2008.02134.x>
- [12] M. Mohieldin and D. Ratha, “Financial inclusion and the SDGs,” in *The Migration and Remittances Factbook*, World Bank, 2017.
- [13] R. Auer, C. Boar, G. Cornelli et al., “CBDCs in emerging market economies,” *BIS Papers*, no. 123, Bank for International Settlements, 2022. <https://www.bis.org/publ/bppdf/bispap123.htm>